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FILM FORMING METHOD AND FILM FORMING APPARATUS

Abstract

A film forming method includes, sequentially performing: preparing a substrate having a first metal film and a first insulating film formed in different regions of a surface of the substrate; forming a first self-assembled monolayer on a surface of the first metal film selectively with respect to a surface of the first insulating film; forming a first bonding film on both the surface of the first insulating film and a surface of the first self-assembled monolayer using a first molecular bonding agent; removing the first self-assembled monolayer to remove a portion of the first bonding film in contact with the surface of the first self-assembled monolayer; and forming a second metal film on the surface of a remaining portion of the first bonding film and the surface of the first metal film.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application Nos. 2024-034512 and 2024-124656, filed on Mar. 7, 2024 and Jul. 31, 2024, respectively, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a film forming method and a film forming apparatus.

BACKGROUND

[0003] A film forming method described in Patent Document 1 includes forming a TiN film along a recess of an interlayer insulating film, and filling the recess of the interlayer insulating film with a Ru film in this order. The TiN film improves adhesion of the Ru film to the interlayer insulating film.

[0004] A bonding method described in Patent Document 2 bonds a first substrate and a second substrate using a compound. The compound has an OH group or an OH generating group, an azide group, and a triazine ring in one molecule.

PRIOR ART DOCUMENTS

Patent Documents

[0005] Patent Document 1: Japanese Patent Laid-open Publication No. 2022-152438

[0006] Patent Document 2: International Publication No. 2012/043631

SUMMARY

[0007] According to one embodiment of the present disclosure, a film forming method includes, sequentially performing: preparing a substrate having a first metal film and a first insulating film formed in different regions of a surface of the substrate; forming a first self-assembled monolayer on a surface of the first metal film selectively with respect to a surface of the first insulating film; forming a first bonding film on both the surface of the first insulating film and a surface of the first self-assembled monolayer using a first molecular bonding agent; removing the first self-assembled monolayer to remove a portion of the first bonding film in contact with the surface of the first self-assembled monolayer; and forming a second metal film on the surface of a remaining portion of the first bonding film and the surface of the first metal film. The first molecular bonding agent is an organic compound having a first functional group and a second functional group in one molecule. The first functional group is more likely to bond to the first insulating film than the second functional group. The second functional group is more likely to bond to the second metal film than the first functional group.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0008] The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments of the present disclosure, and together with the general description given above and the detailed description of the embodiments given below, serve to explain the principles of the present disclosure.

[0009] FIG. 1 is a flowchart showing a film forming method according to a first reference example. [0010] FIG. 2A is a cross-sectional view showing S101 in FIG. 1. FIG. 2B is a cross-sectional view showing S102 in FIG. 1. FIG. 2C is a cross-sectional view showing S103 in FIG. 1. FIG. 2D is a cross-sectional view showing S104 in FIG. 1.

[0011] FIG. 3 is a flowchart showing a film forming method according to a first embodiment.

[0012] FIG. 4A is a cross-sectional view showing S202 in FIG. 3. FIG. 4B is a cross-sectional view showing S203 in FIG. 3. FIG. 4C is a cross-sectional view showing S204 in FIG. 3. FIG. 4D is a cross-sectional view showing S205 in FIG. 3. FIG. 4E is a cross-sectional view showing S206 in FIG. 3.

[0013] FIG. 5 is a flowchart showing a film forming method according to a second reference example.

[0014] FIG. 6A is a cross-sectional view showing S101 in FIG. 5. FIG. 6B is a cross-sectional view showing S102 in FIG. 5. FIG. 6C is a cross-sectional view showing S103 in FIG. 5. FIG. 6D is a cross-sectional view showing S104 in FIG. 5. FIG. 6E is a cross-sectional view showing S105 in FIG. 5.

[0015] FIG. 7 is a flowchart showing a film forming method according to a second embodiment.

[0016] FIG. 8A is a cross-sectional view showing S202 in FIG. 7. FIG. 8B is a cross-sectional view showing S203 in FIG. 7. FIG. 8C is a cross-sectional view showing S204 in FIG. 7. FIG. 8D is a cross-sectional view showing S205 in FIG. 7. FIG. 8E is a cross-sectional view showing S206 in FIG. 7. FIG. 8F is a cross-sectional view showing S207 in FIG. 7.

[0017] FIG. 9 is a flowchart showing a film forming method according to a third reference example, and is a flowchart of a process performed following S103 in FIG. 5.

[0018] FIG. 10A is a cross-sectional view showing S111 in FIG. 9. FIG. 10B is a cross-sectional view showing S112 in FIG. 9. FIG. 10C is a cross-sectional view showing S113 in FIG. 9. FIG. 10D is a cross-sectional view showing S114 in FIG. 9.

[0019] FIG. 11 is a flowchart showing a film forming method according to a third reference example, and is a flowchart of a process performed following S205 in FIG. 7.

[0020] FIG. 12A is a cross-sectional view showing S211 in FIG. 11. FIG. 12B is a cross-sectional view showing S212 in FIG. 11. FIG. 12C is a cross-sectional view showing S213 in FIG. 11. FIG. 12D is a cross-sectional view showing S214 in FIG. 11. FIG. 12E is a cross-sectional view showing S215 in FIG. 11. FIG. 12F is a cross-sectional view showing S216 in FIG. 11.

[0021] FIG. 13 is a plan view showing a film forming apparatus according to an embodiment.

[0022] FIG. 14 is a cross-sectional view showing an example of a first processing part shown in FIG. 13.

DETAILED DESCRIPTION

[0023] Reference will now be made in detail to various embodiments, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure.

However, it will be apparent to one of ordinary skill in the art that the present disclosure may be practiced without these specific details. In other instances, well-known methods, procedures, systems, and components have not been described in detail so as not to unnecessarily obscure aspects of the various embodiments.

[0024] Hereinafter, embodiments of the present disclosure will be described with reference to the drawings. Throughout the drawings, the same or similar components are designated by the same reference numerals, and descriptions thereof may be omitted.

[0025] A film forming method according to a first reference example will be described with reference to FIGS. 1 and 2A to 2D. In the film forming method according to the first reference example, a metal wiring is formed by a dual damascene method. As shown in FIG. 1, the film forming method according to the first reference example includes steps S101 to S104. The film forming method may include steps other than steps S101 to S104. For example, the film forming

method may include a step of cleaning a substrate surface between steps **S101** and **S102**.

[0026] Step **S101** includes preparing a substrate **1** as shown in FIG. 2A. The substrate **1** includes a semiconductor substrate or a glass substrate (not shown), and various functional films formed on the semiconductor substrate or the glass substrate. The substrate **1** includes a first metal film **11** and a first insulating film **12** in different regions of a substrate surface **1a**. The first metal film **11** is, for example, a Ru film. The first metal film **11** may include Ru and may be a Ru alloy film. The first metal film **11** may be a Cu film, a Co film, a W film, or a Mo film. The first metal film **11** may be an alloy film containing at least one of Cu, Co, W, or Mo. The first insulating film **12** is, for example, an interlayer insulating film.

[0027] The interlayer insulating film is preferably a SiO film or a low dielectric constant (low-k) film having a dielectric constant lower than that of the SiO film. The low-k film is not particularly limited, but may be, for example, a SiCO film, a SiOCH film, or a SiCN film. In this regard, the SiO film means a film containing silicon (Si) and oxygen (O). The atomic ratio of the elements constituting the SiO film is not limited to a stoichiometric ratio. Similarly, the SiCO film, the SiOCH film, and the SiCN film contain respective elements, and the atomic ratio of the respective elements constituting each film is not limited to a stoichiometric ratio.

[0028] The substrate **1** has a recess **1b** on the substrate surface **1a**. The recess **1b** has, for example, a trench **1c** and a via hole **1d** formed in the bottom surface of the trench **1c**. The substrate **1** has a surface of a first metal film **11** on a bottom surface of the recess **1b**, and a surface of a first insulating film **12** on a side surface of the recess **1b**. The substrate **1** has a top surface of a protrusion **1e** on the substrate surface **1a**, and a recess **1b** recessed from the top surface of the protrusion **1e**. The substrate **1** has a surface of the first insulating film **12** on the top surface of the protrusion **1e**.

[0029] The substrate **1** may further include a second insulating film **13**. The second insulating film **13** is, for example, an interlayer insulating film. In a recess of the second insulating film **13**, a first metal film **11** and a third adhesive film **14** are filled in this order. The third adhesive film **14** improves adhesiveness between the first metal film **11** and an etching stopper film **15** described later. The third adhesive film **14** is, for example, a TiN film or a TaN film. The atomic ratio of the elements constituting the TiN film or the TaN film is not limited to a stoichiometric ratio.

[0030] The substrate **1** may further include an etching stopper film **15**. The etching stopper film **15** is formed on the second insulating film **13** and the third adhesive film **14**. The etching stopper film **15** stops the etching of the first insulating film **12** for formation of the recess **1b**. The etching stopper film **15** is, for example, a SiN film or a SiCN film. The atomic ratio of elements constituting the SiN film or the SiCN film is not limited to a stoichiometric ratio.

[0031] After the etching of the first insulating film **12** for forming the recess **1b** is completed, a part of the etching stopper film **15** and a part of the third adhesive film **14** are removed to expose the first metal film **11** on the bottom surface of the recess **1b** (see FIG. 2A). As a result, the substrate **1** may have a surface of the third adhesive film **14** and a surface of the etching stopper film **15** on the side surface of the recess **1b**.

[0032] Step **S102** includes forming a first adhesive film **16** as shown in FIG. 2B. The first adhesive film **16** is formed over the entire substrate surface **1a** conformally with the shape of the substrate surface **1a**. The first adhesive film **16** improves adhesiveness between the first insulating film **12** and a second metal film **17**, which will be described later. The first adhesive film **16** is, for example, a TiN film or a TaN film. The first adhesive film **16** is formed by, for example, a CVD (Chemical Vapor Deposition) method or an ALD (Atomic Layer Deposition) method.

[0033] Step **S103** includes forming a second metal film **17** as shown in FIG. 2C. The second metal film **17** is, for example, a Ru film. The second metal film **17** may include Ru and may be a Ru alloy film. The second metal film **17** may be a Cu film, a Co film, a W film, or a Mo film. The second metal film **17** may be an alloy film containing at least one of Cu, Co, W, or Mo. The second metal film **17** is formed by, for example, a PVD (Physical Vapor Deposition) method, a CVD method, or

a plating method.

[0034] The second metal film **17** fills the recess **1b**. The second metal film **17** contacts the first adhesive film **16** at the bottom surface of the recess **1b**. The second metal film **17** also contacts the first adhesive film **16** at the side surface of the recess **1b** and the top surface of the protrusion **1e**. The first adhesive film **16** improves the adhesiveness between the first insulating film **12** and the second metal film **17**.

[0035] Step **S104** includes polishing the second metal film **17** as shown in FIG. 2D. The polishing method is, for example, CMP (Chemical Mechanical Polishing). Step **S104** includes polishing the second metal film **17**, and then simultaneously polishing the second metal film **17** and the first insulating film **12**. A surface of the second metal film **17** and the surface of the first insulating film **12** are aligned in the same plane. The second metal film **17** filled in the first insulating film **12** may form a metal wiring.

[0036] According to the first reference example, as shown in FIG. 2D, the first adhesive film **16** remains between the second metal film **17** and the first metal film **11**. The remaining first adhesive film **16** increases contact resistance between the second metal film **17** and the first metal film **11**. Therefore, in the first embodiment described later, a first bonding film is used instead of the first adhesive film **16**, and a first self-assembled monolayer is formed before the formation of the first bonding film. Hereinafter, the first self-assembled monolayer may be referred to as a first SAM (Self-Assembled Monolayer).

[0037] A film forming method according to a first embodiment will be described with reference to FIGS. 3 and 4A to 4E. In the film forming method according to the first embodiment, a metal wiring is formed by a dual damascene method. As shown in FIG. 3, the film forming method according to the first embodiment includes steps **S201** to **S206**. The film forming method may include steps other than steps **S201** to **S206**. For example, the film forming method may include a step of cleaning a substrate surface between steps **S201** and **S202**.

[0038] Step **S201** includes preparing a substrate **1**. The substrate **1** to be prepared is similar to the substrate **1** prepared in **S101** of the first reference example (see FIG. 2A), and therefore the illustration and description thereof will be omitted.

[0039] Step **S202** includes forming a first SAM **18** as shown in FIG. 4A. The material of the first SAM **18** is, for example, an organic compound containing a thiol group. Specific examples of the thiol-based compound include CF.sub.3(CF.sub.2).sub.5CH.sub.2CH.sub.2SH (1H,1H,2H,2H-perfluorooctanethiol: PFOT) and CH.sub.3(CH.sub.2).sub.5SH (hexanethiol: HT).

[0040] The thiol-based compound is more likely to be chemically adsorbed onto the surface of the first metal film **11** than onto the surface of the first insulating film **12**. Therefore, the first SAM **18** is formed on the surface of the first metal film **11** selectively with respect to the surface of the first insulating film **12**. The first SAM **18** is hardly formed on the surface of the first insulating film **12**.

[0041] The material of the first SAM **18** is not limited to the thiol-based compound. The raw material gas of the first SAM **18** may include a phosphonic acid-based compound, a carboxylic acid-based compound, or a nitro-based compound. The material of the first SAM **18** may be supplied in a liquid state dissolved in an organic solvent, or in a state of a gas vaporized by heating.

[0042] Step **S203** includes forming a first bonding film **19** as shown in FIG. 4B. The first bonding film **19** is formed on both the surface of the first insulating film **12** and a surface of the first SAM **18**. The first bonding film **19** may be formed over the entire substrate surface **1a** conformally with a shape of the substrate surface **1a**. The first bonding film **19** may also be formed on the surface of the third adhesive film **14** and the surface of the etching stopper film **15** inside the recess **1b**. The material of the first bonding film **19** is a first molecular bonding agent. The first molecular bonding agent may be supplied in a liquid state dissolved in an organic solvent, or in a state of a gas vaporized by heating.

[0043] The first molecular bonding agent is an organic compound having a first functional group and a second functional group in one molecule. The first functional group is more likely to be

bonded to the first insulating film **12** than the second functional group. The first functional group is not particularly limited, but may include at least one of a silanol group or a group that generates a silanol group by hydrolysis reaction. The group that generates a silanol group by hydrolysis reaction is, for example, an alkoxysilyl group. A plurality of first functional groups may be present in one molecule, and the plurality of first functional groups may be the same or different.

[0044] The second functional group is more likely to be bonded to the second metal film **17** than the first functional group. The second functional group is not particularly limited, but may include, for example, at least one of an amino group, an azide group, a mercapto group, an isocyanate group, a ureido group, or an epoxy group. A plurality of second functional groups may be present in one molecule, and the plurality of second functional groups may be the same or different.

[0045] The first molecular bonding agent preferably has a triazine between the first functional group and the second functional group. The first molecular bonding agent may contain at least one of benzene, cyclohexane, naphthalene, spiro compound, lactone, pyridine, cyclopentane, furan, or thiophene, instead of the triazine.

[0046] As shown in FIG. **4C**, step **S204** includes removing the first SAM **18** to remove the portion of the first bonding film **19** that contacts the surface of the first SAM **18**. The remaining portion of the first bonding film **19** covers the surface of the first insulating film **12**, and the surface of the first metal film **11** is exposed again at the bottom of the recess **1b**.

[0047] Removing the first SAM **18** includes, for example, heating the first SAM **18**. The first SAM **18** may be thermally decomposed. The heating temperature is not particularly limited as long as it is higher than a 1% weight loss temperature of the first SAM **18**, and is, for example, 300 degrees C. to 450 degrees C. The 1% weight loss temperature is a temperature at which the weight of a substance decreases by 1% when the substance is heated from the room temperature at a heating rate of 10 degrees C./min in a nitrogen atmosphere. The 1% weight loss temperature is measured by a differential thermal analyzer. The holding time of the heating temperature is set according to the heating temperature, and is, for example, 1 minute to 60 minutes, preferably 1 minute to 30 minutes.

[0048] The 1% weight loss temperature of the first SAM **18** is preferably lower than the 1% weight loss temperature of the first bonding film **19**. Even if the heating temperature is a low temperature at which the thermal decomposition of the first bonding film **19** hardly progresses, a portion of the first bonding film **19** may be removed. The heating temperature may be lower than the 1% weight loss temperature of the first bonding film **19**.

[0049] Furthermore, removing the first SAM **18** may include supplying acetic acid to the first SAM **18**. The acetic acid breaks the bond between the first SAM **18** and the first metal film **11**, and removes the first SAM **18**. The acetic acid breaks, for example, a sulfide bond. Therefore, supplying the acetic acid is particularly effective when the first SAM **18** is a thiol-based compound. The supply time of the acetic acid is not particularly limited, but may be, for example, 1 minute to 60 minutes, preferably 1 minute to 30 minutes.

[0050] The supply of the acetic acid to the first SAM **18** is preferably performed after the heating of the first SAM **18**, but may be performed before the heating of the first SAM **18**. Further, step **S204** may include only one of the heating the first SAM **18** and the supplying the acetic acid to the first SAM **18**.

[0051] Step **S205** includes forming a second metal film **17** as shown in FIG. **4D**. The second metal film **17** is, for example, a Ru film. The second metal film **17** may include Ru and may be a Ru alloy film. The second metal film **17** may be a Cu film, a Co film, a W film, or a Mo film. The second metal film **17** may also be an alloy film containing at least one of Cu, Co, W, or Mo. The second metal film **17** is formed by, for example, a CVD method or a plating method.

[0052] The second metal film **17** fills the recess **1b**. The second metal film **17** contacts the first metal film **11** at the bottom surface of the recess **1b**. There is no other functional film between the first metal film **11** and the second metal film **17**. The second metal film **17** contacts the first

bonding film **19** at the side surface of the recess **1b** and at the top surface of the protrusion **1e**. The first bonding film **19** bonds the first insulating film **12** and the second metal film **17**. This makes it possible to suppress the peeling of the second metal film **17** from the first insulating film **12**. The first bonding film **19** may bond the etching stopper film **15** and the second metal film **17**.

[0053] Step **S206** includes polishing the second metal film **17** as shown in FIG. **4E**. The polishing method is, for example, CMP. Step **S206** includes polishing the second metal film **17** and then simultaneously polishing the second metal film **17** and the first insulating film **12**. The surface of the second metal film **17** and the surface of the first insulating film **12** are aligned in the same plane. The second metal film **17** filled in the first insulating film **12** may form a metal wiring.

[0054] According to the first embodiment, as shown in FIG. **4E**, there is no other functional film between the second metal film **17** and the first metal film **11**. The second metal film **17** and the first metal film **11** are in continuous contact with each other. Therefore, the contact resistance between the second metal film **17** and the first metal film **11** may be reduced. In addition, according to the first embodiment, the adhesiveness between the first insulating film **12** and the second metal film **17** may be improved by the remaining portion of the first bonding film **19**.

[0055] A film forming method according to a second reference example will be described with reference to FIGS. **5** and **6A** to **6E**. In the film forming method according to the second reference example, a metal wiring is formed by a semi-damascene method. The film forming method according to the second reference example includes steps **S101** to **S105** as shown in FIG. **5**. The following mainly describes the differences from the first reference example.

[0056] Step **S101** includes preparing a substrate **1** as shown in FIG. **6A**. The substrate **1** has the recess **1b** on the substrate surface **1a**. The recess **1b** has the via hole **1d** shown in FIG. **2A**, but does not have the trench **1c** shown in FIG. **2B**. The substrate **1** has a surface of a first metal film **11** on the bottom surface of the recess **1b**, and a surface of a first insulating film **12** on the side surface of the recess **1b**. The substrate **1** has a top surface of a protrusion **1e** on the substrate surface **1a**, and a recess **1b** recessed from the top surface of the protrusion **1e**. The substrate **1** has a surface of a first insulating film **12** on the top surface of the protrusion **1e**.

[0057] Step **S102** includes forming a first adhesive film **16** as shown in FIG. **6B**. The first adhesive film **16** is formed over the entire substrate surface **1a** conformally with the shape of the substrate surface **1a**.

[0058] Step **S103** includes forming a second metal film **17** as shown in FIG. **6C**. The second metal film **17** fills the recess **1b**. The second metal film **17** contacts the first adhesive film **16** at the bottom surface of the recess **1b**. The second metal film **17** also contacts the first adhesive film **16** at the side surface of the recess **1b** and at the top surface of the protrusion **1e**. The first adhesive film **16** improves the adhesiveness between the first insulating film **12** and the second metal film **17**.

[0059] Step **S104** includes polishing the second metal film **17** as shown in FIG. **6D**. The second metal film **17** is planarized. The second metal film **17** covers the first insulating film **12**, and the surface of the first insulating film **12** is not exposed.

[0060] Step **S105** includes patterning the second metal film **17** as shown in FIG. **6E**. The patterning may be performed using, for example, a photolithography technique and an etching technique. The remaining second metal film **17** may form a metal wiring.

[0061] According to the second reference example, as shown in FIG. **6E**, the first adhesive film **16** remains between the second metal film **17** and the first metal film **11**. The remaining first adhesive film **16** increases the contact resistance between the second metal film **17** and the first metal film **11**. Therefore, in a second embodiment described later, a first bonding film is used instead of the first adhesive film **16**, and a first self-assembled monolayer is formed before the formation of the first bonding film.

[0062] A film forming method according to a second embodiment will be described with reference to FIGS. **7** and **8A** to **8F**. In the film forming method according to the second embodiment, a metal wiring is formed by a semi-damascene method. The film forming method according to the second

embodiment includes steps **S201** to **S207** as shown in FIG. 7. The following mainly describes the differences from the first embodiment.

[0063] Step **S201** includes preparing a substrate **1**. The substrate **1** to be prepared is similar to the substrate **1** prepared in **S101** of the second reference example (see FIG. 6A), and therefore the illustration and description thereof will be omitted.

[0064] Step **S202** includes forming a first SAM **18** as shown in FIG. 8A. The first SAM **18** is formed on the surface of the first metal film **11** selectively with respect to the surface of the first insulating film **12**. The first SAM **18** is hardly formed on the surface of the first insulating film **12**.

[0065] Step **S203** includes forming a first bonding film **19** as shown in FIG. 8B. The first bonding film **19** is formed on both the surface of the first insulating film **12** and the surface of the first SAM **18**. The material of the first bonding film **19** is the first molecular bonding agent.

[0066] Step **S204** includes removing the first SAM **18** as shown in FIG. 8C to remove the portion of the first bonding film **19** that contacts the surface of the first SAM **18**. The remaining portion of the first bonding film **19** covers the surface of the first insulating film **12**, and the surface of the first metal film **11** is exposed again at the bottom of the recess **1b**.

[0067] Step **S205** includes forming a second metal film **17** as shown in FIG. 8D. The second metal film **17** fills the recess **1b**. The second metal film **17** contacts the first metal film **11** at the bottom surface of the recess **1b**. There is no other functional film between the first metal film **11** and the second metal film **17**. The second metal film **17** contacts the first bonding film **19** at the side surface of the recess **1b** and at the top surface of the protrusion **1e**. The first bonding film **19** bonds the first insulating film **12** and the second metal film **17**. This may suppress peeling of the second metal film **17** from the first insulating film **12**. The first bonding film **19** may bond the etching stopper film **15** and the second metal film **17**.

[0068] Step **S206** includes polishing the second metal film **17** as shown in FIG. 8E. The second metal film **17** is planarized. The second metal film **17** covers the first insulating film **12**, and the surface of the first insulating film **12** is not exposed.

[0069] Step **S207** includes patterning the second metal film **17** as shown in FIG. 8F by using, for example, a photolithography technique and an etching technique. The remaining second metal film **17** may form a metal wiring.

[0070] According to the second embodiment, as shown in FIG. 8F, there is no other functional film between the second metal film **17** and the first metal film **11**. The second metal film **17** and the first metal film **11** are in continuous contact with each other. Therefore, the contact resistance between the second metal film **17** and the first metal film **11** may be reduced. In addition, according to the second embodiment, the adhesiveness between the first insulating film **12** and the second metal film **17** may be improved by the remaining portion of the first bonding film **19**.

[0071] A film forming method according to a third reference example will be described with reference to FIGS. 9 and 10A to 10D. The film forming method according to the third reference example forms a metal wiring by a semi-damascene method different from that of the second reference example. As shown in FIG. 9, the film forming method according to the third reference example includes steps **S111** to **S114** following step **S103** shown in FIG. 5. The following mainly describes the differences from the second reference example.

[0072] As shown in FIG. 10A, step **S111** includes polishing the second metal film **17**. Step **S111** includes polishing the second metal film **17** and then simultaneously polishing the second metal film **17** and the first insulating film **12**. The surface of the second metal film **17** and the surface of the first insulating film **12** are aligned in the same plane.

[0073] Step **S112** includes forming a second adhesive film **22** as shown in FIG. 10B. The second adhesive film **22** is formed on both the surface of the second metal film **17** and the surface of the first insulating film **12**. The second adhesive film **22** improves the adhesiveness between the first insulating film **12** and a third metal film **23** described below. The second adhesive film **22** is, for example, a TiN film or a TaN film. The second adhesive film **22** is formed by, for example, a CVD

method or an ALD method.

[0074] Step **S113** includes forming a third metal film **23** as shown in FIG. **10C**. The third metal film **23** contacts the second adhesive film **22**. The second adhesive film **22** improves adhesiveness between the first insulating film **12** and the third metal film **23**. The third metal film **23** is, for example, a Ru film. The third metal film **23** may include Ru, and may be a Ru alloy film. The third metal film **23** may be a Cu film, a Co film, a W film, or a Mo film. The third metal film **23** may also be an alloy film containing at least one of Cu, Co, W, or Mo. The third metal film **23** is formed by, for example, a CVD method or a plating method.

[0075] Step **S114** includes patterning the third metal film **23** as shown in FIG. **10D**. The patterning may be performed using, for example, a photolithography technique and an etching technique. The remaining third metal film **23** may form a metal wiring.

[0076] According to the third reference example, as shown in FIG. **10D**, the second adhesive film **22** remains between the third metal film **23** and the second metal film **17**. The remaining second adhesive film **22** increases contact resistance between the third metal film **23** and the second metal film **17**. Therefore, in a third embodiment described later, a second bonding film is used instead of the second adhesive film **22**, and a second self-assembled monolayer is formed before the formation of the second bonding film. Hereinafter, the second self-assembled monolayer may be referred to as a second SAM (Self-Assembled Monolayer).

[0077] A film forming method according to a third embodiment will be described with reference [0078] to FIGS. **11** and **12A** to **12F**. The film forming method according to the third embodiment forms a metal wiring by a semi-damascene method different from that of the second embodiment. As shown in FIG. **11**, the film forming method according to the third embodiment includes steps **S211** to **S216** following step **S205** shown in FIG. **7**. The following mainly describes the differences from the second embodiment.

[0079] As shown in FIG. **12A**, step **S211** includes polishing the second metal film **17**. Step **S211** includes polishing the second metal film **17** and then simultaneously polishing the second metal film **17** and the first insulating film **12**. The surface of the second metal film **17** and the surface of the first insulating film **12** are aligned in the same plane.

[0080] Step **S212** includes forming a second SAM **25** as shown in FIG. **12B**. The material of the second SAM **25** is similar to the material of the first SAM **18**, and therefore the description thereof will be omitted. The second SAM **25** is formed on the surface of the second metal film **17** selectively with respect to the surface of the first insulating film **12**. The second SAM **25** is hardly formed on the surface of the first insulating film **12**.

[0081] Step **S213** includes forming a second bonding film **26** as shown in FIG. **12C**. The second bonding film **26** is formed on both the surface of the first insulating film **12** and the surface of the second SAM **25**. The material of the second bonding film **26** is a second molecular bonding agent. The second molecular bonding agent may be supplied in a liquid state dissolved in an organic solvent, or in a state of a gas vaporized by heating.

[0082] The second molecular bonding agent is an organic compound having a third functional group and a fourth functional group in one molecule. The third functional group is more likely to bond to the first insulating film **12** than the fourth functional group. The third functional group is not particularly limited, but may include at least one of a silanol group or a group that generates a silanol group by hydrolysis reaction. The group that generates a silanol group by hydrolysis reaction is, for example, an alkoxysilyl group. A plurality of third functional groups may be present in one molecule, and the plurality of third functional groups may be the same or different.

[0083] The fourth functional group is more likely to bond to the third metal film **23** than the third functional group. The fourth functional group is not particularly limited, but may include, for example, at least one of an amino group, an azide group, a mercapto group, an isocyanate group, a ureido group, or an epoxy group. A plurality of fourth functional groups may be present in one molecule, and the plurality of fourth functional groups may be the same or different.

[0084] The second molecular bonding agent preferably includes a triazine between the third and fourth functional groups. The second molecular bonding agent may include at least one of benzene, cyclohexane, naphthalene, spiro compound, lactone, pyridine, cyclopentane, furan, or thiophene, instead of triazine.

[0085] Step S214 includes removing the second SAM 25 shown in FIG. 12D to remove the portion of the second bonding film 26 that contacts the surface of the second SAM 25. The remaining portion of the second bonding film 26 covers the surface of the first insulating film 12, and the surface of the second metal film 17 is exposed again.

[0086] Removing the second SAM 25 includes, for example, heating the second SAM 25. The second SAM 25 may be thermally decomposed. The heating temperature is not particularly limited as long as it is higher than a 1% weight loss temperature of the second SAM 25, and is, for example, 300 degrees C. to 450 degrees C. The holding time of the heating temperature is set according to the heating temperature, and is, for example, 1 minute to 60 minutes, preferably 1 minute to 30 minutes.

[0087] The 1% weight loss temperature of the second SAM 25 is preferably lower than the 1% weight loss temperature of the second bonding film 26. Even if the heating temperature is low enough that the thermal decomposition of the second bonding film 26 hardly progresses, a portion of the second bonding film 26 may be removed. The heating temperature may be lower than the 1% weight loss temperature of the second bonding film 26.

[0088] Further, removing the second SAM 25 may include supplying acetic acid to the second SAM 25. The acetic acid breaks the bond between the second SAM 25 and the second metal film 17, and removes the second SAM 25. The acetic acid breaks, for example, sulfide bonds. Therefore, supplying acetic acid is particularly effective when the second SAM 25 is a thiol-based compound. The supply time of acetic acid is not particularly limited, but may be, for example, 1 minute to 60 minutes, preferably 1 minute to 30 minutes.

[0089] The supply of acetic acid to the second SAM 25 is preferably performed after the heating of the second SAM 25, but may be performed before the heating of the second SAM 25. Further, step S214 may include only one of the heating the second SAM 25 and the supplying the acetic acid to the second SAM 25.

[0090] Step S215 includes forming the third metal film 23 as shown in FIG. 12E. The third metal film 23 is, for example, a Ru film. The third metal film 23 may include Ru and may be a Ru alloy film. The third metal film 23 may be a Cu film, a Co film, a W film, or a Mo film. The third metal film 23 may also be an alloy film containing at least one of Cu, Co, W, or Mo. The third metal film 23 is formed by, for example, a CVD method or a plating method.

[0091] The third metal film 23 is in contact with the second metal film 17. There is no other functional film between the second metal film 17 and the third metal film 23. In addition, the third metal film 23 is in contact with the remaining portion of the second bonding film 26. The remaining portion of the second bonding film 26 bonds the first insulating film 12 and the third metal film 23. This makes it possible to suppress the peeling of the third metal film 23 from the first insulating film 12.

[0092] Step S216 includes patterning the third metal film 23 as shown in FIG. 12F by using, for example, a photolithography technique and an etching technique. The remaining third metal film 23 may form a metal wiring.

[0093] According to the third embodiment, as shown in FIG. 12F, there is no other functional film between the third metal film 23 and the second metal film 17. The third metal film 23 and the second metal film 17 are in continuous contact with each other. Therefore, the contact resistance between the third metal film 23 and the second metal film 17 may be reduced. In addition, according to the third embodiment, the adhesiveness between the first insulating film 12 and the third metal film 23 may be improved by the remaining portion of the second bonding film 26.

[0094] The film forming method may include preparing a substrate 1A (see FIG. 12A) obtained in

step **S211** of the third embodiment. The substrate **1A** has no recess on a substrate surface **1Aa**. The substrate surface **1Aa** is a planar surface on which the surface of the second metal film **17** and the surface of the first insulating film **12** are aligned.

[0095] In this case, the second metal film **17**, the second SAM **25**, and the second bonding film **26** may correspond to the first metal film, the first SAM, and the first bonding film recited in the claims. Accordingly, in this case, the second bonding agent, the third functional group, and the fourth functional group may correspond to the first bonding agent, the first functional group, and the second functional group recited in the claims.

[0096] Next, a film forming apparatus **100** for carrying out the above-mentioned film forming method will be described with reference to FIG. **13**. The film forming apparatus **100** carries out the film forming method shown in FIG. **3**, for example. The film forming apparatus **100** may carry out the film forming method shown in FIG. **7** or FIG. **11**. As shown in FIG. **13**, the film forming apparatus **100** includes a first processing part **200A**, a second processing part **200B**, a third processing part **200C**, a transfer part **400**, and a control circuit **500**. The first processing part **200A** carries out, for example, step **S202** in FIG. **3**. The second processing part **200B** carries out, for example, steps **S203** to **S204** in FIG. **3**. The third processing part **200C** carries out, for example, step **S205** in FIG. **3**. The processing from step **S206** onwards is performed outside the film forming apparatus **100**, but may also be performed inside the film forming apparatus **100**. The first processing part **200A**, the second processing part **200B**, and the third processing part **200C** may have the same structure or may have different structures. It is also possible to perform all of steps **S202** to **S205** in FIG. **3** only with the first processing part **200A**. The transfer part **400** transfers the substrate **1** to the first processing part **200A**, the second processing part **200B**, and the third processing part **200C**. The control circuit **500** controls the first processing part **200A**, the second processing part **200B**, the third processing part **200C**, and the transfer part **400**.

[0097] The transfer part **400** includes a first transfer chamber **401** and a first transfer mechanism **402**. The internal atmosphere of the first transfer chamber **401** is an air atmosphere. The first transfer mechanism **402** is installed inside the first transfer chamber **401**. A load port **405** is installed on a wall surface of the first transfer chamber **401**. A carrier **C** in which a substrate **1** is accommodated is attached to the load port **405**. For example, a FOUP (Front Opening Unified Pod) or the like may be used as the carrier **C**. The first transfer mechanism **402** includes an arm **403** that holds the substrate **1** and travels along a rail **404**. The rail **404** extends in the arrangement direction of the carrier **C**.

[0099] The transfer part **400** also includes a second transfer chamber **411** and a second transfer mechanism **412**. The internal atmosphere of the second transfer chamber **411** is a vacuum atmosphere. The second transfer mechanism **412** is installed inside the second transfer chamber **411**. The second transfer mechanism **412** includes an arm **413** that holds the substrate **1**. The arm **413** is arranged to be movable in the vertical and horizontal directions and rotatable about a vertical axis. The first processing part **200A**, the second processing part **200B**, and the third processing part **200C** are connected to the second transfer chamber **411** via different gate valves **G**.

[0100] Further, the transfer part **400** includes a load lock chamber **421** between the first transfer chamber **401** and the second transfer chamber **411**. The internal atmosphere of the load lock chamber **421** may be switched between a vacuum atmosphere and an air atmosphere by a pressure regulation mechanism (not shown). This allows the inside of the second transfer chamber **411** to be constantly maintained in a vacuum atmosphere. In addition, it is possible to prevent a gas from flowing from the first transfer chamber **401** into the second transfer chamber **411**. Gate valves **G** are provided between the first transfer chamber **401** and the load lock chamber **421**, and between the second transfer chamber **411** and the load lock chamber **421**.

[0101] The control circuit **500** is, for example, a computer, and includes an arithmetic part **501** such as a CPU (Central Processing Unit) or the like and a memory part **502** such as a memory or the like. The memory part **502** stores programs that control various processes executed in the film

forming apparatus **100**. The control circuit **500** controls the operation of the film forming apparatus **100** by causing the arithmetic part **501** to execute the programs stored in the memory part **502**. The control circuit **500** controls the first processing part **200A**, the second processing part **200B**, the third processing part **200C**, and the transfer part **400** to carry out the above-mentioned film forming method.

[0102] The control circuit **500** includes an electronic circuit such as a CPU, an FPGA (Field Programmable Gate Array) or an ASIC (Application Specific Integrated Circuit), and performs the various control operations described in this specification by executing instruction codes stored in a memory or by using a circuit designed for a specific application.

[0103] Next, the operation of the film forming apparatus **100** will be described. First, the first transfer mechanism **402** takes out the substrate **1** from the carrier C, transfers the taken-out substrate **1** to the load lock chamber **421**, and retreats from the load lock chamber **421**. Next, the internal atmosphere of the load lock chamber **421** is switched from the air atmosphere to a vacuum atmosphere. Thereafter, the second transfer mechanism **412** takes out the substrate **1** from the load lock chamber **421**, and transfers the taken-out substrate **1** to the first processing part **200A**.

[0104] Next, the first processing part **200A** performs step S202. Thereafter, the second transfer mechanism **412** takes out the substrate **1** from the first processing part **200A** and transfers the taken-out substrate **1** to the second processing part **200B**. During this time, the atmosphere around the substrate **1** may be maintained as a vacuum atmosphere, which makes it possible to suppress unintended oxidation of the substrate **1**.

[0105] Next, the second processing part **200B** performs steps S203 to S204. Thereafter, the second transfer mechanism **412** takes out the substrate **1** from the second processing part **200B** and transfers the taken-out substrate **1** to the third processing part **200C**. During this time, the atmosphere around the substrate **1** may be maintained as a vacuum atmosphere, which makes it possible to suppress unintended oxidation of the substrate **1**.

[0106] Next, the third processing part **200C** performs step S205. Thereafter, the second transfer mechanism **412** takes out the substrate **1** from the third processing part **200C**, transfers the taken-out substrate **1** to the load lock chamber **421**, and retreats from the load lock chamber **421**. The internal atmosphere of the load lock chamber **421** is then switched from the vacuum atmosphere to an air atmosphere. Thereafter, the first transfer mechanism **402** takes out the substrate **1** from the load lock chamber **421**, and stores the taken-out substrate **1** in the carrier C. Then, the processing of the substrate **1** is completed.

[0107] Next, the first processing part **200A** will be described with reference to FIG. **14**. The second processing part **200B** and the third processing part **200C** are configured similarly to the first processing part **200A**, and thus, the illustration and description will be omitted.

[0108] The first processing part **200A** includes a substantially cylindrical airtight processing container **210**. An exhaust chamber **211** is provided at the center of the bottom wall of the processing container **210**. The exhaust chamber **211** has, for example, a substantially cylindrical shape that protrudes downward. An exhaust pipe **212** is connected to the exhaust chamber **211**, for example, at a side surface of the exhaust chamber **211**.

[0109] An exhaust source **272** is connected to the exhaust pipe **212** via a pressure controller **271**. The pressure controller **271** includes a pressure regulation valve such as a butterfly valve or the like. The exhaust pipe **212** is configured so that the inside of the processing container **210** may be depressurized by the exhaust source **272**. The pressure controller **271** and the exhaust source **272** constitute a gas exhaust mechanism **270** that exhausts a gas from the inside of the processing container **210**.

[0110] A transfer port **215** is provided on the side surface of the processing container **210**. The transfer port **215** is opened and closed by a gate valve G. The substrate **1** is transferred between the processing container **210** and the second transfer chamber **411** (see FIG. **13**) via the transfer port **215**.

[0111] A stage **220**, which is a holder for holding the substrate **1**, is provided in the processing container **210**. The stage **220** holds the substrate **1** horizontally with the substrate surface **1a** facing upward. The stage **220** is formed in a substantially circular shape in a plan view, and is supported by a support member **221**. A substantially circular recess **222** for holding the substrate **1** having a diameter of, for example, 300 mm is formed on the surface of the stage **220**. The recess **222** has an inner diameter slightly larger than the diameter of the substrate **1**. The depth of the recess **222** is set to be, for example, substantially the same as a thickness of the substrate **1**. The stage **220** is formed of a ceramic material such as aluminum nitride (AlN) or the like. The stage **220** may also be formed of a metallic material such as nickel (Ni) or the like. Instead of the recess **222**, a guide ring for guiding the substrate **1** may be provided on the peripheral edge portion of the surface of the stage **220**.

[0112] A grounded lower electrode **223** is embedded in the stage **220**. A heating mechanism **224** is embedded below the lower electrode **223**. The heating mechanism **224** heats the substrate **1** held on the stage **220** to a set temperature by being supplied with electric power from a power supplier (not shown) based on a control signal from the control circuit **500** (see FIG. **13**). When the entire stage **220** is made of a metal, the entire stage **220** functions as a lower electrode. Therefore, the lower electrode **223** does not need to be embedded in the stage **220**.

[0113] The stage **220** is provided with a plurality of (e.g., three) lift pins **231** for holding and lifting the substrate **1** held on the stage **220**. The material of the lift pins **231** may be, for example, ceramics such as alumina (Al.sub.2O.sub.3) or quartz. The lower ends of the lift pins **231** are attached to a support plate **232**. The support plate **232** is connected to a lifting mechanism **234** provided outside the processing container **210** via a lifting shaft **233**.

[0114] The lifting mechanism **234** is installed, for example, at the bottom of the exhaust chamber **211**. A bellows **235** is installed between the lifting mechanism **234** and an opening **219** for the lifting shaft **233** formed at the bottom surface of the exhaust chamber **211**. The support plate **232** may be shaped so that it may be raised and lowered without interfering with the support member **221** of the stage **220**. The lifting pins **231** are configured to be freely raised and lowered by the lifting mechanism **234** between above the surface of the stage **220** and below the surface of the stage **220**.

[0115] A gas supplier **240** is provided at the ceiling wall **217** of the processing container **210** via an insulating member **218**. The gas supplier **240** forms an upper electrode and faces the lower electrode **223**. A radio-frequency power source **252** is connected to the gas supplier **240** via a matcher **251**. By supplying radio-frequency power of 450 kHz to 100 MHz from the radio-frequency power source **252** to the upper electrode (gas supplier **240**), a radio-frequency electric field is generated between the upper electrode (gas supplier **240**) and the lower electrode **223**, thereby generating capacitively coupled plasma. A plasma generator **250** that generates plasma includes the matcher **251** and the radio-frequency power source **252**. The plasma generator **250** may generate not only the capacitively coupled plasma, but also other plasma such as inductively coupled plasma. In a process that does not generate plasma, the gas supplier **240** does not need to form an upper electrode, and the lower electrode **223** is also not needed.

[0116] The gas supplier **240** includes a hollow gas supply chamber **241**. A number of holes **242** for dispersing and supplying a processing gas into the processing container **210** are arranged, for example, uniformly on the bottom surface of the gas supply chamber **241**. A heating mechanism **243** is embedded in the gas supplier **240**, for example, above the gas supply chamber **241**. The heating mechanism **243** is heated to a set temperature by being supplied with electric power from a power supplier (not shown) based on a control signal from the control circuit **500**.

[0117] A gas supply mechanism **260** is connected to the gas supply chamber **241** via a gas supply path **261**. The gas supply mechanism **260** supplies gases used in the film forming method to the gas supply chamber **241** via the gas supply path **261**. Although not shown, the gas supply mechanism **260** includes an individual pipe for each type of gas, an opening/closing valve installed in the

middle of the individual pipe, and a flow rate controller installed in the middle of the individual pipe. When the opening/closing valve opens the individual pipe, a gas is supplied from the supply source to the gas supply path **261**. The supply amount of the gas is controlled by the flow rate controller. On the other hand, when the opening/closing valve closes the individual pipe, the supply of the gas from the supply source to the gas supply path **261** is stopped.

[0118] Although the embodiments of the film forming method and the film forming apparatus according to the present disclosure have been described above, the present disclosure is not limited to the above-described embodiments. Various changes, modifications, substitutions, additions, deletions, and combinations may be made within the scope recited in the claims. It goes without saying that these also fall within the technical scope of the present disclosure.

[0119] According to the present disclosure in some embodiments, it is possible to reduce electrical resistance between a first metal film and a second metal film and to improve adhesiveness between a first insulating film and the second metal film.

[0120] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosures. Indeed, the embodiments described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions, and changes in the form of the embodiments described herein may be made without departing from the spirit of the disclosures. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosures.

Claims

1. A film forming method, comprising, sequentially performing: preparing a substrate having a first metal film and a first insulating film formed in different regions of a surface of the substrate; forming a first self-assembled monolayer on a surface of the first metal film selectively with respect to a surface of the first insulating film; forming a first bonding film on both the surface of the first insulating film and a surface of the first self-assembled monolayer using a first molecular bonding agent; removing the first self-assembled monolayer to remove a portion of the first bonding film in contact with the surface of the first self-assembled monolayer; and forming a second metal film on a surface of a remaining portion of the first metal film and the surface of the first insulating film, wherein the first molecular bonding agent is an organic compound having a first functional group and a second functional group in one molecule, wherein the first functional group is more likely to bond to the first insulating film than the second functional group, and wherein the second functional group is more likely to bond to the second metal film than the first functional group.
2. The method of claim 1, wherein the removing the first self-assembled monolayer includes heating the first self-assembled monolayer.
3. The method of claim 2, wherein a 1% weight loss temperature of the first self-assembled monolayer is lower than a 1% weight loss temperature of the first bonding film.
4. The method of claim 1, wherein the removing the first self-assembled monolayer includes supplying acetic acid to the first self-assembled monolayer.
5. The method of claim 1, wherein a material of the first self-assembled monolayer is an organic compound containing a thiol group.
6. The method of claim 1, wherein the first metal film and the second metal film contain Ru.
7. The method of claim 1, wherein the substrate has a recess on the surface of the substrate, wherein a bottom surface of the recess includes the surface of the first metal film, and wherein a side surface of the recess includes the surface of the first insulating film.
8. The method of claim 1, wherein the first functional group includes at least one of a silanol group or a group that generates the silanol group by hydrolysis reaction.

9. The method of claim 1, wherein the second functional group includes at least one of an amino group, an azide group, a mercapto group, an isocyanate group, a ureido group, or an epoxy group.

10. The method of claim 1, wherein the first molecular bonding agent includes a triazine between the first functional group and the second functional group.

11. The method of claim 1, further comprising, sequentially performing: polishing the second metal film to align a surface of the second metal film and the surface of the first insulating film in the same plane; forming a second self-assembled monolayer on the surface of the second metal film selectively with respect to the surface of the first insulating film; forming a second bonding film on both the surface of the first insulating film and a surface of the second self-assembled monolayer using a second molecular bonding agent; removing the second self-assembled monolayer to remove a portion of the second bonding film in contact with the surface of the second self-assembled monolayer; and forming a third metal film on a surface of a remaining portion of the second bonding film and the surface of the second metal film, wherein the second molecular bonding agent is an organic compound having a third functional group and a fourth functional group in one molecule, wherein the third functional group is more likely to bond to the first insulating film than the fourth functional group, and wherein the fourth functional group is more likely to bond to the third metal film than the third functional group.

12. A film forming apparatus, comprising: a processing container; a holder configured to hold a substrate inside the processing container; a gas supply mechanism configured to supply a gas into the processing container; a gas exhaust mechanism configured to exhaust a gas from the processing container; a transfer mechanism configured to load and unload the substrate into and from the processing container; and a control circuit configured to control the gas supply mechanism, the gas exhaust mechanism, and the transfer mechanism to carry out the film forming method of claim 1.
